PROJECT HANDOVER / 28 OCTOBER 2026 / MUMBAI

Leave Management AWS Deployment Guide

Member 3: Approvals, Notifications and Audit

A practical record of the code, cloud resources, deployment decisions, demo accounts and verification completed for this project.

Deployment	Value
AWS account / region	773384831067 / ap-south-1 (Mumbai)
CloudFormation stack	leave-management-system-aws-v2
Backend runtime	All four Lambda functions: Java 17 on Amazon Linux 2023
API stage / RDS database	stage1 / lms_approval

Deployed links

Open the frontend
https://leave-management-system-frontend-773384831067-ap-south-1.s3.ap-south-1.amazonaws.com/index.html

Check backend health: /ping
https://96h1em86gl.execute-api.ap-south-1.amazonaws.com/stage1/ping

Backend base URL
https://96h1em86gl.execute-api.ap-south-1.amazonaws.com/stage1

Login accounts

Manager: manager (employee ID 100). Retrieve manager password.

Demo employee: employee.demo (employee ID 900100). Retrieve employee password.

> Open the linked secret in AWS Secrets Manager, choose **Retrieve secret value**, and copy the **password** value. Passwords and bearer tokens are intentionally absent from this PDF.

Guide sections: scope and workflow; Lambda names; AWS service names; networking; backend; API reference; Postman; demo creation; build/deploy; verification; maintenance and cleanup.

01 / APPLICATION SCOPE

What was built

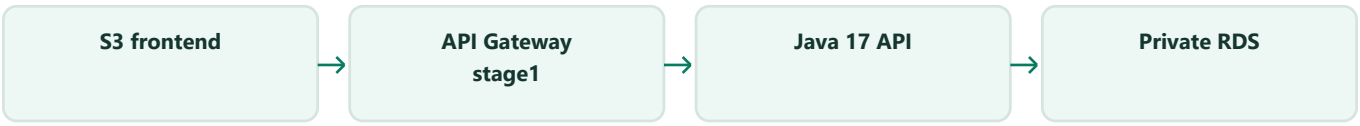
The previous project was used as the source for the required Member 3 backend and selected frontend components. The original project was left unchanged.

Location	Path
Original project	E:\eclipse-workspace\Leave-Management-System
AWS project	E:\eclipse-workspace\leave-management-system-aws

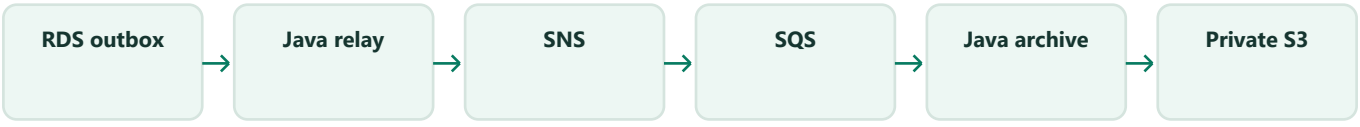
Requirement	Implemented behavior
Approve or reject	Only an authenticated MANAGER can decide a PENDING request.
Check assignment	The acting manager must match the employee manager_id. Self-approval is rejected.
Record decision	Approval row stores leave ID, manager ID, decision, comment and time. A comment of 1-200 characters is required.
Deduct balance	Approval increments used days and reduces remaining days. Rejection does not deduct days.
Notify employee	Creates an owned in-app notification and an outbox event in the same database transaction.
Keep audit history	Records affected employee, leave, acting manager, action and comment.
Support login	Provisioned accounts, BCrypt passwords and expiring database-backed bearer sessions. No public registration.

How one decision travels

HTTP REQUEST / DATABASE TRANSACTION



COMMITTED EVENT DELIVERY / RETRIES



The status, balance, approval, notification, audit and outbox changes commit together. A failed transaction rolls them all back. The scheduled outbox relay sends committed events later, so an SNS outage does not undo a completed approval.

Delivery is at least once. The event UUID determines the S3 key; repeated delivery writes the same logical key. S3 versioning may keep multiple object versions.

02 / LAMBDA INVENTORY

Four functions, four responsibilities

AWS displays **Java 17 on Amazon Linux 2023**. Java 17 is the language runtime; Amazon Linux is the operating system underneath it. No EC2 virtual machine was deployed for this application.

AWS function name	Use / invocation
leave-management-system-lambda-aws	Main Spring Boot API: login, assigned pending queue, approve/reject, notifications and audit. Invoked by API Gateway.
leave-management-system-outbox-aws	Reads up to 10 unpublished outbox rows, publishes to SNS and marks delivery. EventBridge invokes it every minute.
leave-management-system-archive-aws	Consumes raw SNS JSON delivered through SQS, validates event UUIDs and writes audit/events/<eventId>.json into private S3.
leave-management-system-bootstrap-aws	Initializes schema, restricted app grants and manager account. The approved demo update also creates the demo employee and leave. Invoked by CloudFormation.

Handlers and settings

Function	Handler class / resources
Main API	com.nexturn.lms.StreamLambdaHandler::handleRequest 1024 MB; 30-second timeout
Outbox	com.nexturn.lms.aws.OutboxLambdaHandler::handleRequest 512 MB; 60-second timeout
Archive	com.nexturn.lms.aws.ArchiveLambdaHandler::handleRequest 512 MB; 30-second timeout
Bootstrap	com.nexturn.lms.aws.SchemaBootstrapHandler::handleRequest 1024 MB; 180-second timeout

The archive helper was initially Python 3.13. At your request it was replaced with Java 17. The final deployment uses runtime identifier **java17.al2023** for all four functions. No archive role or bucket access was broadened.

The bootstrap is an infrastructure helper rather than an API endpoint. It is retained so future CloudFormation schema updates have their service token available.

03 / AWS NAMES AND USES

Application and data services

AWS resource name / identifier	Purpose
leave-management-system-api-aws **API ID: 96h1em86gl** **Stage: stage1**	Regional REST API. ANY /{proxy+} forwards requests to the main Lambda. The application enforces bearer authentication.
leave-management-system-rds-aws	RDS MySQL instance: db.t4g.micro, Single-AZ, 20 GiB encrypted gp3. Private database lms_approval; one-day automated backups.
leave-management-system-notifications-aws	Encrypted SNS topic carrying committed leave-decision events. The configured subscriber is the project SQS queue; no email or SMS subscription.
leave-management-system-events-aws	SQS work queue for the archive Lambda. 180-second visibility timeout; 14-day retention; batch size 10.
leave-management-system-events-dlq-aws	SQS dead-letter queue. Messages move here after five failed receives. Retention: 14 days.
leave-management-system-frontend-773384831067-ap-south-1	S3 bucket for compiled index.html, JavaScript, CSS and config.json. Public GetObject over HTTPS only; no public list or write access.
leave-management-system-aws-773384831067-ap-south-1	Private versioned S3 bucket for deployment ZIPs/templates and archived audit/events/ JSON objects.
leave-management-system-aws-v2	CloudFormation stack managing the network, database, functions and event/API connections. The artifact bucket was created separately.

RDS endpoint

leave-management-system-rds-aws.c5mc0o4u4um1.ap-south-1.rds.amazonaws.com

The RDS endpoint is private and is not a browser link. Use the HTTPS API for application requests. Lambda connects to MySQL on port 3306 with hostname and certificate verification.

04 / AWS NAMES AND USES

Network, credentials and access

Name / CloudFormation logical ID	Use
Vpc: vpc-054ba38c266bf8be6	Dedicated 10.42.0.0/16 network. No NAT or internet gateway in this project VPC.
SubnetA: subnet-0cd38d6d9a94adcac **SubnetB: subnet-070599507b4014673**	Private 10.42.0.0/24 and 10.42.1.0/24 subnets in separate availability zones. Used by RDS and database-connected Lambdas.
RouteTableA / RouteTableB **RouteAssociationA / RouteAssociationB**	Separate subnet routes. The S3 gateway endpoint supplies the required S3 route.
LambdaSecurityGroup **sg-03fbe06b811bc816f**	Outbound MySQL/TLS connections from the API, bootstrap and outbox functions.
DatabaseSecurityGroup **sg-0abf853ef0d7f5b6d**	MySQL inbound only from the project Lambda security group.
EndpointSecurityGroup **sg-0f2c7065bbc7ae7d2**	TLS inbound to the SNS endpoint only from the project Lambda group.
S3Endpoint: vpce-05c431efe7c262b4b **SnsEndpoint: vpce-0af213d84c7c096a5**	Private access to S3 and SNS. The SNS endpoint policy permits the outbox role to publish to this topic.
ApiRole / BootstrapRole **OutboxRole / ArchiveRole**	Four generated IAM execution roles. API/bootstrap: logs and VPC access. Outbox: scoped SNS/KMS access. Archive: SQS receive/delete and S3 audit-prefix writes.
OutboxSchedule / SchedulePermission	EventBridge rate(1 minute) and permission to invoke the outbox function. Physical rule name is generated by CloudFormation.

Secrets Manager names

Secret name	Use
DatabaseAdminSecret-hSyhyXdpvCRV	lms_admin database credential, used by schema initialization.
DatabaseAppSecret-RZMh2yNaU7Ij	Restricted lms_app database credential, used by API and outbox.
InitialManagerSecret-HgyoWZFuAaYc	manager application login; employee ID 100.
leave-management-system-demo-employee-aws	employee.demo application login; employee ID 900100.

05 / SOURCE GUIDE

Backend files and data protection

File / component	Responsibility
Application.java / StreamLambdaHandler.java	Local Spring Boot startup and API Gateway REST event adapter.
ManagerController / ApprovalServiceImpl	Pending queue, assignment and status checks, balance locks, transactional decision.
LoginController / LoginService	BCrypt verification, token issue/authentication, current identity and logout.
NotificationController / NotificationServiceImpl	Create, list and mark the current employee notifications as read.
AuditLogController / AuditLogService	Append decision history; permit employee or their assigned manager to read it.
GlobalExceptionHandler / ApiException	Central JSON errors. Java source has no explicit try/catch blocks.
aws/SchemaBootstrapHandler.java	Idempotent schema setup, restricted database grants and approved demo seed.
aws/ArchiveLambdaHandler.java	Java 17 SQS-to-S3 archive, canonical UUID validation and batch retry on failure.
aws/OutboxLambdaHandler.java	Scheduled publish using database locks and a separate delivery transaction.
deployment/cloudformation.json	Complete infrastructure definition; use this for the deployed new-RDS setup.

Tables used

Shared module data: employee, leave_type, leave_application, leave_balance. **Member 3 data:** approval, notification, audit_log, outbox_event. **Authentication:** login and auth_session.

The lms_app user can only select shared employee/leave-type records. It can update leave status/balances and insert required decision records. Administrative schema changes are limited to bootstrap credentials. Passwords use BCrypt; session records contain SHA-256 token digests rather than raw bearer tokens.

Decision and balance rows use write locks; employee manager assignment uses a shared lock compatible with SELECT-only permissions. Requests spanning balance years must be split upstream. Stored positive working-day counts are supplied by the leave module.

The full file-by-file descriptions are in docs/FILE-GUIDE.md. Global errors contain timestamp, status, message and path; responses do not expose stack traces.

06 / API REFERENCE

Routes and request bodies

All paths below are relative to the deployed base URL on page 1. Use JSON for request bodies and **Authorization: Bearer <accessToken>** for protected routes.

Method and route	Use
GET /ping	Public health check. Returns 200 and {"status":"UP"}.
POST /api/auth/login	Login with username/password; returns accessToken, tokenType, expiresAt, username, empId and role.
GET /api/auth/me	Get the authenticated identity.
POST /api/auth/logout	Revoke the current session.
GET /api/manager/leaves/pending	Manager-only assigned pending requests; supports page and size.
POST /api/manager/leave/{leaveId}/approve	Approve with a required comment; deducts balance.
POST /api/manager/leave/{leaveId}/reject	Reject with a required comment; no balance deduction.
GET /api/notifications	Current employee notifications only; paginated.
PATCH /api/notifications/{id}/read	Mark an owned notification as read.
GET /api/audit-logs/employee/{empId}	Own history or, for managers, an assigned employee history.

Login body

```
{
  "username": "manager",
  "password": "PASTE_FROM_SECRETS_MANAGER"
}
```

Approval or rejection body

```
{"comment": "Approved after review"}
```

Do not send managerId or employee identity in the decision body. The server gets the actor from the bearer token. Lists default to 20 records; the maximum page size is 100.

07 / POSTMAN WALKTHROUGH

Test manager and employee flows

1. Create a collection and set **baseUrl** to the deployed API base URL. Set **leaveId** from the pending response, not from a guessed number. Set **employeeId** to **900100** for the demo.

2. Send **GET {{baseUrl}}/ping**. Expected: HTTP 200 and status UP.

3. Send **POST {{baseUrl}}/api/auth/login**. Select Body > raw > JSON. Use username manager and its Secrets Manager password. Copy accessToken from the response.

4. For manager calls, select Authorization > Bearer Token and paste that token. Send **GET {{baseUrl}}/api/manager/leaves/pending?page=0&size=20**. The demo request should appear if it is still pending.

5. Copy its leaveId. Send **POST {{baseUrl}}/api/manager/leave/{{leaveId}}/approve** with the JSON comment from page 7. Expected: APPROVED. The demo remaining balance changes from 20 to 18 days.

6. Log in separately as **employee.demo** with the demo employee secret. Use its own token for **GET {{baseUrl}}/api/notifications** and **GET {{baseUrl}}/api/audit-logs/employee/900100**. Copy an owned notification ID and PATCH its /read route.

7. To test rejection, use /reject on a different assigned pending request. An already decided request is not reset by the demo initializer.

Optional Postman login response script

In the login request Tests / Post-response script, save the token to a collection variable:

```
const result = pm.response.json();
if (pm.response.code === 200) {
  pm.collectionVariables.set("accessToken", result.accessToken);
}
```

Then use **{{accessToken}}** in the Bearer Token field. Logging in as another account replaces that variable; use the correct actor token for each check. A companion importable collection is supplied without passwords.

Check	Expected result
No bearer token	401
Employee calls manager route	403
Empty decision comment	400
Approve an already decided leave	409
Valid assigned pending decision	200

Approve/reject requests change real database state. Run them individually against the labelled demo record; do not run both decisions on the same leave as if both should succeed.

08 / DEMO ACCOUNT CREATION

Exactly how the employee was added

Initially the deployment contained only manager 100. You then approved adding one synthetic demo employee and a test leave. The seed runs through the existing Java 17 CloudFormation database initializer; no employee-management module or public registration endpoint was added.

Record	Created value
Employee	emp_id 900100; name Demo Employee; manager_id 100; synthetic address; no real personal details.
Login	username employee.demo; role EMPLOYEE; linked to employee 900100; BCrypt password hash.
Leave type	leave_type_id 900001; Demo Annual Leave; quota 20; negative balance disabled.
Balance	20 allocated, 0 used, 20 remaining for the generated leave year.
Pending request	Two requested days, generated future dates within one year; reason Demo request for Member 3 approval verification.
Credential secret	leave-management-system-demo-employee-aws; AWS generates the password; it is not stored in source or this guide.

Creation sequence

• CloudFormation creates DemoEmployeeSecret using GenerateSecretString with username employee.demo and a 32-character generated password.

• The secret is resolved into the bootstrap environment as DEMO_EMPLOYEE_PASSWORD. DEMO_DATA_ENABLED is set to true.

• SchemaInitialization uses version 2-demo-employee so CloudFormation invokes the updated initializer.

• The initializer inserts the employee and checks that the ID belongs to the labelled demo employee assigned to manager 100.

• It creates the employee login only if absent, hashes the password with BCrypt, and verifies the username belongs to employee 900100 with EMPLOYEE role.

• It inserts the demo leave type and yearly balance if missing, then inserts one test request only if that labelled request does not already exist.

The seed is idempotent: repeating it does not duplicate the demo leave or reset an approved request or used balance. It does not create genuine staff records. Use your employee/leave modules for real operational data.

09 / BUILD AND DEPLOYMENT

Local build and AWS update steps

Backend

Use a Java 17 JDK and Maven. From the AWS project directory:

```
mvn clean verify -Plambda
```

The Lambda ZIP is generated under target/. The code targets Java 17 (class-file major 61). During this work, tests also ran on an actual Temurin Java 17 runtime; a newer installed JDK compiled with --release 17.

Frontend

```
cd frontend
npm ci
npm run build
```

frontend/public/config.json contains the public API URL; the build copies it to dist. Upload the **contents** of dist to the frontend bucket root: index.html, the compiled JS and CSS files, and config.json. Do not upload node_modules or a frontend ZIP. Hash routing supports nested pages on static hosting.

CloudFormation deployment procedure

• Build and test first. Put the Lambda ZIP and matching template into the private artifact bucket using unique artifact keys.

• Open leave-management-system-aws-v2 in CloudFormation, choose Update stack and replace the template using its private S3 URL.

• Keep existing artifact/secret parameters. Review the change preview, including IAM capabilities and any resource replacement.

• Submit the reviewed update and monitor Events. Wait for UPDATE_COMPLETE; an uploaded ZIP alone is not a deployment.

• Read Outputs for ApiBaseUrl, FrontendUrl, RdsInstanceName and secret ARNs. Update/upload frontend config.json if the API URL changes.

• Verify health, protected-route authentication, CORS, the manager pending queue and SNS-to-SQS-to-S3 delivery.

Final artifact keys

Main API/outbox: backend-828346eb2db07607.zip
Updated archive/bootstrap: archive-java17-3d4a404dbbf3b1ec.zip
Final template: cloudformation-all-java17-3d4a404dbbf3b1ec.json

The archive conversion reuses the same function name, queue, event source mapping, role and bucket prefix. It updates code/runtime in place rather than creating a fifth function.

10 / VERIFICATION

Checks completed and troubleshooting

Verification	Observed result
Java build and tests	31 enabled tests passed; two opt-in MySQL tests skipped in the final build. The earlier restricted-MySQL test passed separately.
Archive tests	Five local real-SDK HTTP tests: JSON/encryption write, stable key, UUID rejection, S3 failure propagation and empty batch.
Demo initializer	Actual Java 17 initializer run twice against isolated MySQL: one demo account/request, correct manager, BCrypt login and 20-day balance.
Deployed infrastructure	CloudFormation update complete; four Java 17 functions; private project RDS available.
API and browser	Health 200/UP; unauthenticated manager route 401; CORS accepts exact frontend origin; hosted login renders.
Cloud archive	Non-sensitive SNS event traversed SQS and the Java archive worker and appeared in private S3.
Demo data in cloud	Manager and employee.demo logins passed in the deployed browser. Manager queue shows request 1 for employee 900100, 29-30 October 2026, two days, still PENDING.

Troubleshooting

Symptom	Check
Manager list is empty	Log in as manager. Confirm demo seed completion and employee manager_id 100. If already decided, the request correctly disappears.
401 / 403	401: login again or use the current token. 403: use the correct role and assigned manager/owned employee.
409 conflict	Refresh status; check balance, leave year and whether another decision already completed.
Cannot reach service	Check /ping, frontend config.json and exact CORS origin. The API base must include /stage1.
Archive missing	Check outbox published_at, scheduled relay errors, SNS subscription, SQS/DLQ and archive Lambda logs.
Initial request is slow	Java cold start after inactivity can add latency. Inspect CloudWatch for failures before retrying a decision.

CloudWatch log groups: /aws/lambda/ followed by each exact function name on page 3. Log retention is 14 days. Do not log passwords, bearer tokens or full secret values.

11 / DEPLOYMENT RECORD

Recovery decisions, cleanup and maintenance

Event	Action and result
Java version	Initial Java 21 preparation was replaced by a tested Java 17 backend at your request. The archive helper was later converted from Python to Java 17.
VPC quota reached	First stack rolled back. You confirmed deletion of unused vpc-045d07c90f37c0db7, three empty subnets and its internet gateway. Dedicated project networking then succeeded.
CloudFront verification block	AWS refused a new distribution until account verification. You approved S3 HTTPS public reads for compiled frontend files instead.
RDS backup limit	Seven-day retention was rejected by the free-plan limit. One-day automated backups allowed creation.
Old RDS removal	next26-rds-db and its nexturnapplndb data were permanently deleted after your confirmation, without a final snapshot or retained automated backups.
Old Lambda removal	next26_aws_function was deleted after confirmation. Its old next26-lambda-api, logs and roles were not deleted as part of that function deletion.
Current stack	leave-management-system-aws-v2 is the active deployment. The first leave-management-system-aws record remains ROLLBACK_COMPLETE.

What remains outside the functional module

Employee administration, real leave applications, holidays, HR allocations, registration and calendar pages were excluded. The demo seed is the explicit exception you requested to make the approval flow testable. No email/SMS recipient has been configured.

Two unused CloudFront configuration objects remain in the stack: FrontendHeaders (response-headers policy) and FrontendOriginAccess (origin access control). They are not a working CloudFront distribution and do not serve the frontend. Any future cleanup should be performed through a reviewed stack update.

Maintenance notes

• Monitor RDS, Secrets Manager, the SNS interface endpoint, storage and requests for credit use or charges. These resources are not guaranteed to be free.

• Keep generated credentials private. Refresh Lambda configuration when rotating the referenced secrets; automatic rotation is not configured.

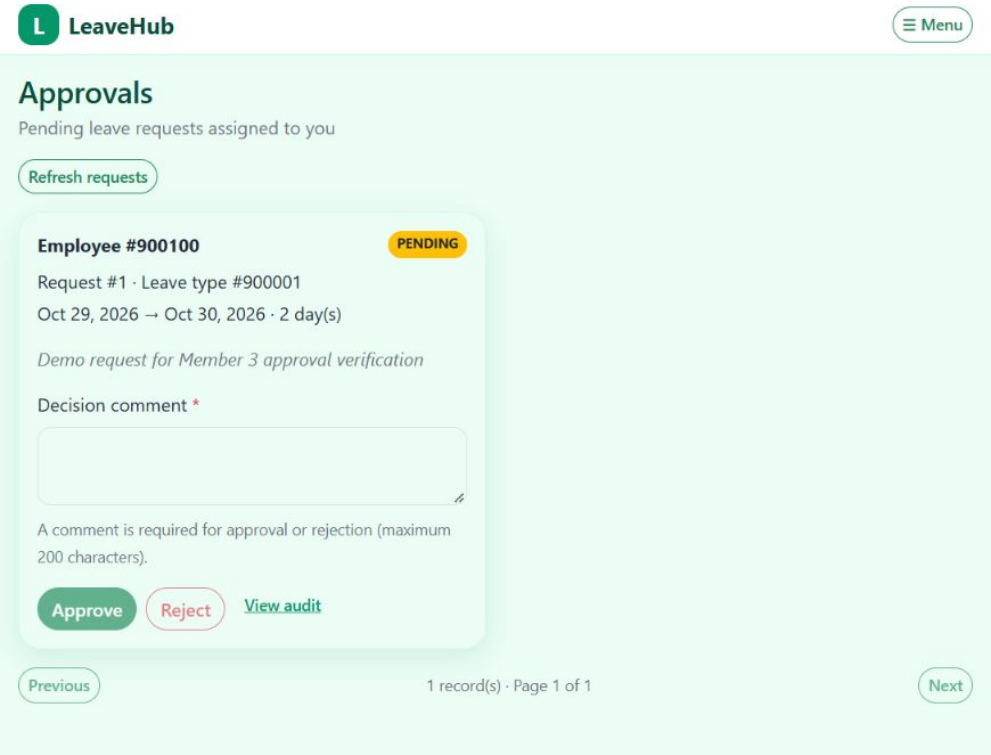
• RDS deletion policy requests a snapshot; frontend bucket and secrets are retained. Deleting the stack does not mean all stored data is erased.

Source references: project README.md; docs/FILE-GUIDE.md; docs/VERIFICATION.md; deployment/cloudformation.json; deployment/STATUS.md; AWS Console deployment and test evidence recorded on 28 October 2026.

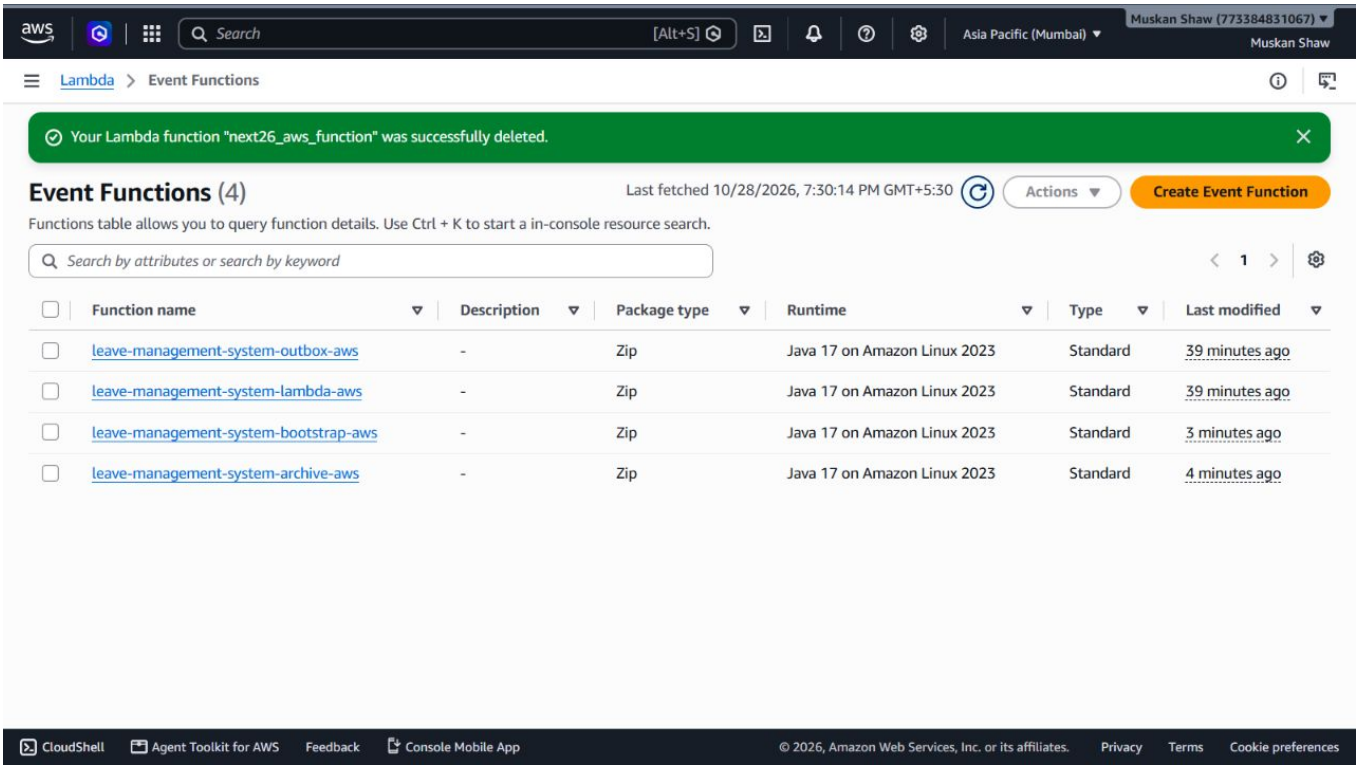
12 / VISUAL EVIDENCE

Deployed interface and runtime inventory

The screenshots below show the deployed application and the final Lambda inventory. They contain no credential values.



Manager pending queue with the synthetic demo employee and request.



Four project functions using Java 17. The obsolete next26 function was removed.